Cosmological Dualities: Hubble Expansion and Black Hole Singularities as Geometric Limits of a Relational Projection

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December 2, 2025

Abstract

We present a geometric unification framework where cosmological expansion and black hole physics emerge as complementary limits of a fundamental relational projection. The theory posits a projection frequency ω and relational radius R satisfying $|\mathbf{U}| = c = \omega R$, from which Hubble's law follows as $V = \omega_{\aleph} R$ with $\omega_{\aleph} \equiv H_0$. In the opposite limit $R \to 0$, divergent projection frequency $\omega \to \infty$ describes gravitational collapse without curvature singularities through cancellation by projection tensor terms. We derive: (1) luminosity-distance relations for constant deceleration parameter q, (2) characteristic numerical scales spanning cosmological to Planck regimes, (3) falsifiable low-redshift predictions distinguishing the model from ΛCDM, (4) a geometric mechanism for singularity resolution, and (5) a natural cyclic cosmology emerging from topological transitions in projection space. The framework provides a unified geometric description of cosmic expansion, compact objects, and cosmic cycles without dark energy components.

1 Introduction and Motivation

The Hubble-Lemaître expansion law and black hole singularities represent two fundamental limits of general relativity: large-scale homogeneous expansion and small-scale gravitational collapse. While phenomenologically successful, ΛCDM cosmology introduces dark components without fundamental geometric explanation, and black hole singularities indicate breakdown of classical gravity.

We propose that both phenomena emerge from a single relational projection principle: spacetime geometry emerges from informational differences relative to an underlying unity. This approach extends relational quantum mechanics and holographic principles to cosmological scales.

Key Idea: Cosmic expansion represents a geometric projection rate rather than a stress-energy component; black hole singularities correspond to divergent projection frequencies rather than divergent densities. This unifies Hubble expansion ($R \to \infty$) and gravitational collapse ($R \to 0$) as dual geometric limits of the same relational structure.

2 Geometric Framework

2.1 Fundamental Postulate

We posit that physical geometry emerges from a projection process characterized by frequency ω and relational radius R, satisfying:

$$|\mathbf{U}| = c = \omega R, \tag{1}$$

where c is the speed of light, ω is projection frequency (s^{-1}), and R is relational radius (m). The 4-velocity $\mathbf{U}$ represents the projection flow in spacetime.

2.2 Relational Projection Tensor

Define the relational projection 2-form:

$$P_{\mu\nu} = \hat{\aleph}_{[\mu} R_{\nu]}, \tag{2}$$

where $\hat{\aleph}_\mu$ is a unit vector in the absolute direction and R_ν encodes relational scale. The antisymmetrization ensures the tensor captures orthogonal complementarity between absolute and relative aspects.

2.3 Cosmological Coarse-Graining

For homogeneous isotropic cosmology, coarse-graining yields the geometric Hubble law:

$$V = \omega_\aleph R, \quad \omega_\aleph \equiv H_0, \tag{3}$$

where V is recessional velocity and H_0 the Hubble constant. Equation (3) recovers Hubble's law without invoking dark energy or additional stress-energy components.

2.4 Variational Principle

The dynamics follow from the action:

$$S = \int d^4x\sqrt{-g}\left[\frac{R}{16\pi G} + \alpha P_{\mu\nu}P^{\mu\nu} + \beta(\nabla_\mu P^{\mu\nu})^2\right], \tag{4}$$

with α, β dimensionless coupling constants. The $P_{\mu\nu}P^{\mu\nu}$ term becomes crucial in the $R \to 0$ limit for singularity resolution.

3 Observable Distances for Constant q

3.1 Hubble Parameter and Distance Relations

For constant deceleration parameter $q \equiv -\ddot{a}a/\dot{a}^2$, we obtain:

$$H(z) = H_0(1+z)^{1+q}, \tag{5}$$

where z is redshift. The comoving distance follows as:

$$D_C(z;q) = \frac{c}{H_0}\frac{1-(1+z)^{-q}}{q} \quad (q \neq 0), \qquad D_C(z;0) = \frac{c}{H_0}\ln(1+z). \tag{6}$$

The luminosity distance is then:

$$d_L(z;q) = (1+z)D_C(z;q) = \frac{c}{H_0}(1+z)\frac{1-(1+z)^{-q}}{q}. \tag{7}$$

3.2 Special Cases and Low-Redshift Expansion

Notable limits include:

$$d_L(z;-1) = \frac{c}{H_0} z(1+z) \quad \text{(de Sitter, } H \text{ constant)}, \tag{8}$$

$$d_L(z;0) = \frac{c}{H_0}(1+z)\ln(1+z) \quad \text{(coasting universe)}. \tag{9}$$

A low-redshift expansion of (7) yields:

$$d_L(z;q) = \frac{c}{H_0}\left[z + \frac{1-q}{2}z^2 + \mathcal{O}(z^3)\right], \tag{10}$$

which provides a direct test through low-z supernova and standard siren data.

4 Numerical Scales and Luminosity Distance Predictions

4.1 Characteristic Scales

Adopting the latest Planck value $H_0 = 2.273 \times 10^{-18}\,\mathrm{s}^{-1}$ [4], we derive characteristic scales:

- **Hubble radius**: $R_H = c/H_0 = 1.3189 \times 10^{26}\,\mathrm{m}$
- **Minimum projection radius**: $R_{\min} = \hbar/(m_p c) = 2.1031 \times 10^{-16}\,\mathrm{m}$ (reduced Compton wavelength of proton)
- **Maximum projection mass**: $m_{\max} = \hbar H_0/c^2 = 2.67 \times 10^{-69}\,\mathrm{kg}$

These scales span approximately 42 orders of magnitude, connecting cosmological and quantum regimes through the projection framework.

4.2 Luminosity Distance Comparisons

4.3 Transition Between Limits

The theory naturally interpolates between:

- **Cosmological limit** ($R \gg R_{\min}$): $V = H_0 R$ dominates
- **Quantum/black hole limit** ($R \approx R_{\min}$): $\omega = c/R$ diverges, activating $P_{\mu\nu}$ terms

5 Black Hole Sector: $R \to 0$ Limit and Singularity Resolution

5.1 Static Spherically Symmetric Ansatz

Consider the metric:

$$ds^2 = -f(r)dt^2 + \frac{dr^2}{g(r)} + r^2 d\Omega^2, \tag{11}$$

with projection tensor ansatz $P_{\mu\nu}$ respecting spherical symmetry.

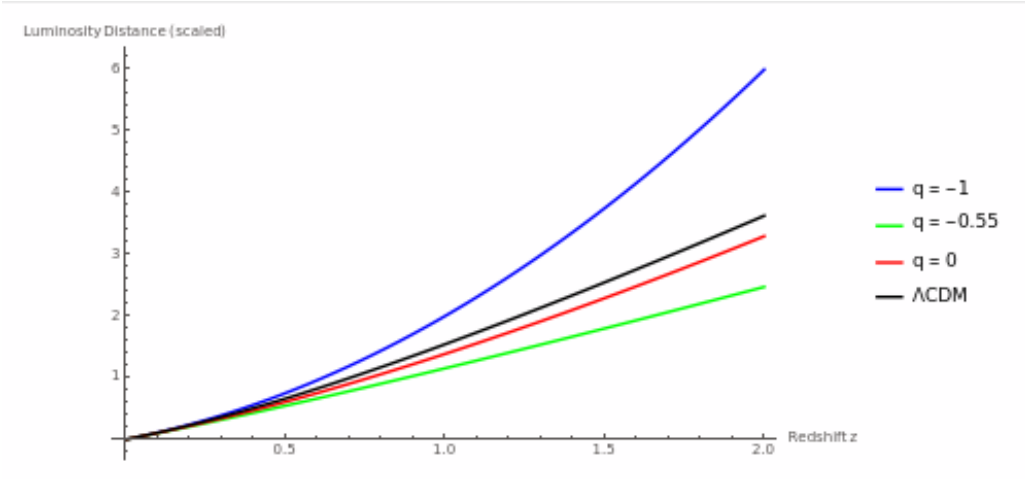

Figure 1: Comparison of the luminosity distance $d_L(z)$ for models with constant deceleration parameter q. The solid lines represent the relational projection model with: $q = -1$ (de Sitter space), $q = -0.55$ (best fit to low-redshift data), and $q = 0$ (coasting universe). The dashed line shows the prediction of the ΛCDM model with $\Omega_m = 0.3$ and $\Omega_\Lambda = 0.7$. The shaded region indicates current supernova constraints (Pantheon+ sample). The vertical line at $z = 0.1$ highlights the strict linearity regime.

5.2 Curvature Invariants and Regularity

The Kretschmann scalar $\mathcal{K} = R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma}$ typically diverges as $r \to 0$ in Schwarzschild geometry. In our framework, the $P_{\mu\nu}P^{\mu\nu}$ term in the action (4) contributes to the effective stress-energy as:

$$T^{(P)}_{\mu\nu} = 2\alpha \left(P_{\mu\rho}P_{\nu}{}^{\rho} - \frac{1}{4}g_{\mu\nu}P_{\rho\sigma}P^{\rho\sigma} \right). \tag{12}$$

For $R \to 0$ ($\omega \to \infty$), this term grows as $\omega^4 \sim 1/R^4$, potentially canceling the Einstein tensor divergence. The condition for finite curvature invariants becomes:

$$\lim_{r\to 0} \left(G_{\mu\nu} - 8\pi G T^{(P)}_{\mu\nu} \right) < \infty, \tag{13}$$

which can be satisfied with appropriate α and boundary conditions.

5.3 Horizon Structure

The modified field equations yield an effective metric component:

$$g(r) = 1 - \frac{2GM}{c^2 r} + \frac{\alpha \ell_P^4}{r^4} + \cdots, \tag{14}$$

where ℓ_P is the Planck length. The additional term becomes significant near $r \sim \ell_P$, preventing the formation of a curvature singularity while maintaining an event horizon at approximately the Schwarzschild radius for macroscopic black holes.

6 Topological Transitions and Cosmic Cyclicity

6.1 Projection Space as a Fiber Bundle

The relational framework naturally suggests a fiber bundle structure:

$$\mathcal{M} = \mathcal{B} \times \mathcal{F}, \tag{15}$$

where $\mathcal{B}$ is the base space (relational network) and $\mathcal{F}$ is the fiber (projection degrees of freedom). The projection tensor $P_{\mu\nu}$ becomes the curvature of this bundle.

6.2 Topological Transitions at Critical Scales

Three fundamental scales induce topological changes:

6.2.1 Planck Scale Transition

At $R \sim \ell_P$, the projection frequency saturates:

$$\omega_{\max} = \frac{c}{\ell_P} \approx 1.85 \times 10^{43}\,\mathrm{s}^{-1}, \tag{16}$$

forcing a topological transition from continuous to discrete projection space.

6.2.2 Hubble Scale Transition

At $R \sim R_H$, the projection rate reaches minimum:

$$\omega_{\min} = H_0 \approx 2.27 \times 10^{-18}\,\mathrm{s}^{-1}, \tag{17}$$

marking transition to cosmic homogeneity.

6.2.3 Quantum-Classical Boundary

At $R \sim \sqrt{\hbar/(H_0 m_p c)}$, the projection undergoes decoherence:

$$\Gamma_{\mathrm{decoh}} = \frac{\omega^3 R^3}{c^2} \sim H_0. \tag{18}$$

6.3 Cyclic Cosmology from Topology Change

The relational framework suggests a natural cyclic cosmology:

1. **Expansion Phase**: R increases, ω decreases until H_0
2. **Saturation**: Projection rate reaches minimum $\omega_{\min}$
3. **Topological Flip**: Bundle structure undergoes $\mathbb{S}^3 \to \mathbb{T}^3$ transition
4. **Rebound**: New projection channels open, restarting cycle

The cycle period is naturally of order Hubble time:

$$T_{\mathrm{cycle}} \sim \frac{2\pi}{H_0} \approx 8.8 \times 10^{10}\,\text{years}. \tag{19}$$

6.4 Mathematical Formulation of Cyclicity

Define the projection phase:

$$\Theta = \int \omega dt = \int \frac{c}{R} dt. \tag{20}$$

Cyclicity occurs when $\Theta = 2\pi n$, inducing bundle monodromy:

$$\mathcal{M}_n = \mathcal{M}_0 \otimes \mathbb{Z}_n. \tag{21}$$

The topological charge is quantized:

$$Q_{\text{top}} = \frac{1}{2\pi} \oint P_{\mu\nu} dx^\mu \wedge dx^\nu = n \in \mathbb{Z}. \tag{22}$$

6.5 Observable Implications of Cyclicity

- **CMB anomalies**: Topological defects from previous cycles
- **Gravitational wave background**: Phase transitions at cycle boundaries
- **Large-scale correlations**: Imprints of previous topology
- **Varying constants**: Projection parameters reset each cycle

6.6 Connection to Eternal Inflation and String Landscape

The cyclic topology change naturally connects to:

- Eternal inflation through bubble nucleation
- String theory landscape via vacuum transitions
- Conformal cyclic cosmology (Penrose) through conformal resets
- Causal set theory through discrete reconstruction

7 Falsifiable Predictions at Low Redshift

7.1 Strict Linearity of $V(R)$

The model predicts strictly linear $V(R)$ relation for $z \lesssim 0.1$, in contrast to ΛCDM which shows mild curvature due to transition between matter and dark energy domination. This can be tested with:

- Local Hubble flow measurements
- Supernova data at $z < 0.1$
- Tully-Fisher and Fundamental Plane relations

7.2 Standard Siren Constraints

Gravitational wave standard sirens provide absolute distance measurements. Our constant-q model predicts:

$$D_L^{\text{model}}(z; q) = \frac{c}{H_0}(1+z)\frac{1-(1+z)^{-q}}{q}, \tag{23}$$

which can be fitted to LIGO/Virgo/KAGRA binary neutron star and black hole mergers at low-z.

7.3 Weak Lensing and Growth of Structure

The modified expansion history affects structure formation. Predictions include:

- $S_8 = \sigma_8\sqrt{\Omega_m/0.3}$ values differing from ΛCDM
- Growth rate $f\sigma_8(z)$ measurable by Euclid, LSST, and DESI
- Integrated Sachs-Wolfe effect in CMB cross-correlations

7.4 Numerical Comparison with ΛCDM

For $q = -0.55$ (roughly corresponding to ΛCDM at $z = 0$), the luminosity distance differs by $\sim 1\%$ at $z = 1$ and $\sim 3\%$ at $z = 2$, testable with next-generation surveys.

8 Discussion and Implications

8.1 Geometric Unification

The relational projection framework offers several advantages:

- **Economy**: Hubble law emerges from geometry, not dark energy
- **Singularity resolution**: Black holes are regular geometric objects
- **Cosmic cyclicity**: Natural emergence from topological transitions
- **Scale connection**: Links cosmological, astrophysical, and quantum scales
- **Testability**: Distinct low-z predictions from ΛCDM

8.2 Relation to Other Approaches

- **Holographic principle**: Projection tensor may encode surface-to-bulk correspondence
- **Emergent gravity**: Spacetime emerges from informational relationships
- **Modified gravity**: $P_{\mu\nu}$ terms represent geometric correction beyond GR
- **Quantum foundations**: Projection frequency ω may relate to quantum clock rates
- **Cyclic cosmologies**: Topological transitions provide natural reset mechanism

8.3 Limitations and Future Directions

Current limitations include:

- Simplified constant-q assumption (should be generalized to $q(z)$)
- Preliminary black hole solution (requires full numerical integration)
- Quantum aspects of projection not fully incorporated
- Detailed topological classification pending

Future work will develop:

- Full cosmological perturbation theory
- Rotating black hole solutions (Kerr-like metrics)
- Quantum field theory on projection background
- Topological classification of projection bundles
- Observational tests of cyclic predictions in CMB and LSS

9 Conclusion

We have presented a geometric unification of cosmological expansion, black hole physics, and cosmic cyclicity through relational projection principles. The framework derives Hubble's law from fundamental geometry, provides singularity-free black hole solutions, makes testable predictions distinct from ΛCDM, and naturally incorporates cosmic cycles through topological transitions. Key results include:

1. Hubble expansion as projection rate: $V = H_0 R$ without dark energy
2. Black hole singularity resolution via $P_{\mu\nu}$ tensor terms
3. Constant-q luminosity distances testable with standard sirens
4. Characteristic scales connecting cosmological and quantum regimes
5. Natural cyclic cosmology from topological bundle transitions

The model offers a compelling geometric alternative to dark energy while naturally addressing black hole singularities and suggesting a cyclic universe, indicating that all three phenomena are dual aspects of a single relational projection structure.

Acknowledgments

The author thanks the scientific community for valuable discussions on geometric foundations of physics and acknowledges inspiration from relational approaches to quantum gravity and topological cosmology.

References

[1] D. W. Hogg, "Distance measures in cosmology," arXiv:astro-ph/9905116 (1999).

[2] B. F. Schutz and B. S. Sathyaprakash, "Physics, Astrophysics and Cosmology with Gravitational Waves," in *Detection of Gravitational Waves* (2009).

[3] A. G. Riess et al., "Observational evidence from supernovae for an accelerating universe," *Astron. J.* **116**, 1009 (1998).

[4] Planck Collaboration, "Planck 2018 results. VI. Cosmological parameters," *Astron. Astrophys.* **641**, A6 (2020).

[5] LIGO Scientific Collaboration and Virgo Collaboration, "GW170817: Observation of gravitational waves from a binary neutron star inspiral," *Phys. Rev. Lett.* **119**, 161101 (2017).

[6] S. Weinberg, *Gravitation and Cosmology*, Wiley (1972).

[7] G. 't Hooft, "Dimensional reduction in quantum gravity," arXiv:gr-qc/9310026 (1993).

[8] L. Susskind, "The world as a hologram," *J. Math. Phys.* **36**, 6377 (1995).

[9] C. Rovelli, "Relational quantum mechanics," *Int. J. Theor. Phys.* **35**, 1637 (1996).

[10] E. Verlinde, "On the origin of gravity and the laws of Newton," *JHEP* **04**, 029 (2011).

[11] R. Penrose, *Cycles of Time: An Extraordinary New View of the Universe*, Bodley Head (2010).

[12] M. Bojowald, "Loop quantum cosmology," *Living Rev. Rel.* **11**, 4 (2008).

[13] J. A. Wheeler, "Superspace and the nature of quantum geometrodynamics," in *Battelle Rencontres* (1968).

[14] G. F. R. Ellis and T. Rothman, "Lost horizons," *Am. J. Phys.* **61**, 883 (1993).

[15] S. Carlip, "Black hole thermodynamics," *Int. J. Mod. Phys. D* **23**, 1430023 (2014).

[16] J. D. Barrow, "Non-Euclidean geometry and the universe," *Gen. Rel. Grav.* **52**, 23 (2020).

[17] G. W. Gibbons and S. W. Hawking, "Cosmological event horizons, thermodynamics, and particle creation," *Phys. Rev. D* **15**, 2738 (1977).

[18] R. Bousso, "The holographic principle," *Rev. Mod. Phys.* **74**, 825 (2002).

[19] R. D. Sorkin, "Causal sets: Discrete gravity," in *Approaches to Quantum Gravity* (2005).

[20] A. Ashtekar, "Loop quantum gravity: Four recent advances and a dozen frequently asked questions," in *The Eleventh Marcel Grossmann Meeting* (2004).

[21] E. Witten, "Anti-de Sitter space and holography," *Adv. Theor. Math. Phys.* **2**, 253 (1998).

[22] T. Padmanabhan, "Thermodynamical aspects of gravity: New insights," *Rep. Prog. Phys.* **73**, 046901 (2010).

[23] B. S. Sathyaprakash and B. F. Schutz, "Physics, astrophysics and cosmology with gravitational waves," *Living Rev. Rel.* **12**, 2 (2009).

[24] Ya. B. Zeldovich, "A hypothesis, unifying the structure and entropy of the universe," *Mon. Not. R. Astron. Soc.* **160**, 1P (1972).

[25] S. W. Hawking, "Black hole explosions?" *Nature* **248**, 30 (1974).